Project Title	**SmartLogix AI: Intelligent Multi-Modal Logistics & Autonomous Delivery Platform using Machine Learning, Computer Vision, LLMs, and AI Agents**
Skills take away From This Project	• Machine Learning Classification • Regression & ETA Prediction • Predictive Maintenance using ML • Route Optimization Techniques • Computer Vision using YOLO • Image Processing with OpenCV • Natural Language Processing (NLP) • Retrieval Augmented Generation (RAG) • LLM Chatbot Development • Multi-Agent AI Systems • Prompt Engineering • SQL Database Design (PostgreSQL/MySQL) • Cloud Storage using AWS S3 • Cloud Deployment using AWS EC2 • Event-Driven Automation using AWS Lambda • Email Notification Systems • REST API Development with FastAPI • Data Visualization & Dashboard Development • System Architecture Design • End-to-End AI Solution Development
Domain	• Artificial Intelligence • Logistics & Supply Chain • E-Commerce • Transportation • Computer Vision • Predictive Analytics • Generative AI • Cloud Computing

Problem Statement:

Modern logistics systems often rely on predefined transportation methods without considering real-time business constraints such as delivery distance, package characteristics, traffic conditions, weather, fleet availability, operational costs, and vehicle health. This frequently results in inefficient resource utilization, delayed deliveries, increased operational expenses, and poor customer experience.

The objective of this project is to develop an AI-powered logistics platform capable of intelligently selecting the most suitable transportation method (Drone, Bike, Van, Truck, Air Cargo, or Ship), optimizing delivery routes, predicting delivery times, monitoring fleet health using Computer Vision and Predictive Analytics, and providing customers with an intelligent LLM-powered conversational assistant for order tracking, product inquiries, and delivery support.

Business Use Cases:

- Intelligent transportation mode selection
- Autonomous drone delivery planning
- Route optimization for multiple deliveries
- Payload optimization for drone operations
- Delivery time (ETA) prediction
- Predictive maintenance for drones and vehicles
- Drone damage detection using Computer Vision
- Automated fleet monitoring
- Real-time order tracking

- AI-powered customer support chatbot
- Product recommendation and comparison assistant
- Customer review sentiment analysis
- Automated delivery notifications
- Delivery analytics dashboard
- Fleet utilization monitoring
- Cost optimization for logistics operations

Approach:

Phase 1: Data Collection

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- Order Information
- Product Details
- Customer Information
- Delivery Locations
- Weather Data
- Traffic Data
- Vehicle Information
- Drone Sensor Data
- Drone Images
- Customer Reviews
- Delivery Logs

Phase 2: Data Engineering

- Data Cleaning
- Feature Engineering
- Missing Value Handling
- Data Transformation
- SQL Database Design
- AWS S3 Image Storage

Phase 3: Machine Learning

Develop multiple AI models:

- Logistics Mode Classification
- Delivery ETA Prediction
- Predictive Maintenance Model

Phase 4: Route Optimization

- Vehicle Routing Problem (VRP)
- Drone Battery Constraints
- Payload Optimization
- Delivery Scheduling

Phase 5: Computer Vision

Train a YOLO model to detect:

- Broken Propeller
- Frame Damage
- Motor Damage
- Landing Gear Damage
- Battery Damage
- Healthy Drone

Phase 6: LLM + RAG

Develop an intelligent chatbot capable of:

- Tracking customer orders
- Answering delivery-related questions
- Comparing products
- Summarizing customer reviews
- Recommending products
- Fetching delivery information from SQL databases
- Answering FAQs using Retrieval-Augmented Generation (RAG)

Phase 7: Cloud Deployment

Deploy the application using:

- AWS EC2
- AWS RDS
- AWS S3
- AWS Lambda
- Email Notification Services

Phase 8: Dashboard & Monitoring

Develop dashboards for:

- Delivery Monitoring
- Fleet Status
- Drone Health
- Customer Orders
- Delivery Analytics

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Results:

The developed platform successfully automates the logistics decision-making process by intelligently selecting the most suitable transportation method based on multiple delivery constraints. It optimizes delivery routes to reduce operational costs, predicts delivery times with improved accuracy, identifies potential fleet maintenance issues before failures occur, and detects drone damages through Computer Vision models.

Additionally, the LLM-powered conversational assistant enables customers to interact naturally with the platform for order tracking, product recommendations, review analysis, and delivery-related queries. The integrated cloud infrastructure provides scalable deployment, automated notifications, centralized data management, and real-time operational monitoring.

Project Evaluation Metrics:

Machine Learning

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

ETA Prediction

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Square Error (RMSE)
- R^2 Score

Computer Vision

- mAP@50
- mAP@50-95
- Precision
- Recall
- IoU

Predictive Maintenance

- Precision
- Recall
- F1-Score
- Confusion Matrix

LLM Chatbot

- Response Accuracy
- Retrieval Precision
- Context Relevance
- User Satisfaction Score

Business Metrics

- Delivery Cost Reduction
- Delivery Time Reduction
- Fleet Utilization
- Drone Utilization Rate
- Customer Satisfaction
- On-Time Delivery Percentage

Technical Tags:

`Machine Learning, Deep Learning, Computer Vision, YOLO, OpenCV, Predictive Maintenance, Route Optimization, Vehicle Routing Problem, ETA Prediction, Drone Analytics, NLP, LLM, Generative AI, RAG, AI Agents, LangChain, FastAPI, Python, PostgreSQL, MySQL, AWS EC2, AWS RDS, AWS S3, AWS Lambda, REST API, SQL, Docker, Streamlit, Data Analytics, Supply Chain Analytics, Logistics Intelligence`

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Data Set: SmartLogix Dataset

The project utilizes multiple datasets to simulate a real-world logistics ecosystem:

- E-Commerce Orders Dataset
- Product Catalog Dataset
- Delivery & Shipment Dataset
- Fleet & Vehicle Dataset
- Drone Flight Sensor Dataset
- Drone Damage Image Dataset
- Customer Reviews Dataset
- Weather Dataset
- Traffic Dataset
- GPS Route Dataset

- Delivery Logs Dataset

Data Set Explanation:

Dataset	Purpose
Customer Orders	Stores order details, delivery priority, package weight, dimensions, and destination information.
Product Catalog	Contains product specifications for chatbot recommendations and comparisons.
Delivery Logs	Maintains shipment history, timestamps, delivery status, and tracking events.
Fleet Information	Stores vehicle, drone, driver, and transportation availability details.
Drone Sensor Data	Includes battery level, motor temperature, flight hours, payload weight, and sensor readings for predictive maintenance.
Drone Images	Used to train the YOLO model for drone damage detection.

Customer Reviews	Enables sentiment analysis and product recommendation capabilities.
Weather Data	Supports transportation mode selection and ETA prediction.
Traffic Data	Assists in route optimization and delivery scheduling.
GPS Coordinates	Used for route optimization and real-time delivery tracking.

Project Deliverables:

- Complete End-to-End AI Logistics Platform
- Machine Learning Classification Model
- ETA Prediction Model
- Predictive Maintenance Model
- Drone Damage Detection Model (YOLO)
- Route Optimization Engine
- AI-Powered LLM Chatbot with RAG
- Multi-Agent Logistics Assistant
- Customer Order Tracking Module
- Product Recommendation System
- Sentiment Analysis Module
- PostgreSQL/MySQL Database
- REST APIs using FastAPI
- AWS Cloud Deployment (EC2, RDS, S3, Lambda)
- Delivery Monitoring Dashboard
- Fleet Management Dashboard
- Technical Documentation
- Source Code Repository
- Model Training Notebooks

- Deployment Guide
- Final Project Presentation

Project Guidelines:

- Design the solution using a modular architecture where each AI component (ML, CV, NLP, LLM, and Optimization) operates independently while integrating seamlessly into the overall workflow.
- Use PostgreSQL or MySQL as the primary relational database and AWS S3 for storing drone inspection images and model artifacts.
- Develop separate Machine Learning models for logistics mode classification, ETA prediction, and predictive maintenance.
- Train a YOLO-based Computer Vision model for drone damage detection using annotated image datasets.
- Implement a Retrieval-Augmented Generation (RAG) chatbot capable of querying SQL databases, delivery logs, product catalogs, and customer purchase history.
- Build AI agents to handle customer support, logistics monitoring, and product recommendation tasks.
- Integrate AWS Lambda to automate delivery notifications and email alerts based on shipment status changes.
- Expose all core functionalities through RESTful APIs using FastAPI to enable frontend and third-party integrations.
- Develop an interactive dashboard to visualize fleet status, delivery performance, drone health, and operational KPIs.
- Evaluate each AI component using appropriate machine learning, computer vision, and business performance metrics, and document the complete development lifecycle from data preparation to cloud deployment.

Timeline:

PROJECT DOUBT CLARIFICATION SESSION (PROJECT AND CLASS DOUBTS)

About Session: The Project Doubt Clarification Session is a helpful resource for resolving questions and concerns about projects and class topics. It provides support in understanding project requirements, addressing code issues, and clarifying class concepts. The session aims to enhance comprehension and provide guidance to overcome challenges effectively.
Note: Book the slot at least before 12:00 Pm on the same day

Timing: Monday-Saturday (4:00PM to 5:00PM)

Booking link :https://forms.gle/XC553oSbMJ2Gcfug9

For DE/BADM project/class topic doubt slot clarification session:

Booking link : https://forms.gle/NtkQ4UV9cBV7Ac3C8

Session timing:

For DE: 04:00 pm to 5:00 pm every saturday
For BADM 05:00 to 07:00 pm every saturday

LIVE EVALUATION SESSION (CAPSTONE AND FINAL PROJECT)

About Session: The Live Evaluation Session for Capstone and Final Projects allows participants to showcase their projects and receive real-time feedback for improvement. It assesses project quality and provides an opportunity for discussion and evaluation.
Note: This form will Open only on Saturday (after 2 PM) and Sunday on Every Week

Timing:

For BADM and DE
Monday-Saturday (11:30AM to 1:00PM)

For DS and AIML
Monday-Saturday (05:30PM to 07:00PM)

Booking link : https://forms.gle/1m2Gsro41fLtZurRA


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